

■ ExcelExportService.java - Full Tanglish Explanation

1■■ @Service Annotation: Ithu Spring Service layer file, meaning — backend logic handle panrathu for Excel export. **2■■ Method Name:** **exportProductsToExcel(List<Product> products)** Ithu product list ah Excel file ah convert panni downloadable format la return panrathu. **3■■ Workbook Creation:** `Workbook workbook = new XSSFWorkbook();` — XSSFWorkbook means `.xlsx` Excel file create pannum. **4■■ Sheet Creation:** `Sheet sheet = workbook.createSheet("Products");` — "Products" nu oru sheet create pannum. **5■■ Header Style & Font:** Header row bold ah irukanum so font bold panna `headerFont.setBold(true);` use panniruka. Cell style apply pannitu header la "ID", "Product Name", etc create panniruka. **6■■ Data Rows Logic:** Loop la ellarum products oda data row by row fill panniruka: Row 0 → Header Row 1, 2... → Product data (ID, Name, Model, etc.) **7■■ Conversion to ByteArray:** `ByteArrayOutputStream` la Excel content write panni `ByteArrayInputStream` ah return panniruka — so controller la download panna mudiyum. **8■■ try-with-resources:** Automatic resource close aagum — Excel workbook & stream close panna manually venam. **■ Summary:** Ithu whole Excel export logic handle pannura class. Database la iruka product data Excel file ah format panni return pannum. Spring service layer la irukura best practice — clean, reusable & memory-safe design.